Literature answer: algebraic reflexivity of $\mathrm{Iso}(C_0(X))$ in the separable case

Automated literature-status note, run `fa_banach_001`

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Status: literature already answered. This note records an exact separate-paper resolution, not a new proof.

Source question

F. Cabello Sánchez and L. Molnár, *Reflexivity of the isometry group of some classical spaces*, arXiv:math/0108056; Rev. Mat. Iberoam. 18 (2002), 409–430.

At the end of the paper (arXiv PDF page 18, journal page 430), the authors ask:

> Let X be a locally compact Hausdorff space. If $C_0(X)$ is separable (equivalently, the one-point compactification of X is metrizable), must $\mathrm{Iso}(C_0(X))$ be algebraically reflexive?

They observe that an affirmative answer would cover every separable commutative C^*-algebra.

Exact supporting theorem

F. Cabello Sánchez, *Local isometries on spaces of continuous functions*, Math. Z. 251 (2005), 735–749, DOI 10.1007/s00209-005-0814-9.

Appendix, Theorem 1 (journal page 748; supporting PDF page 14) states:

> Let L be a locally compact space whose one-point compactification is metrizable. Then $\mathrm{Iso}(C_0^{\mathbb{C}}(L))$ is algebraically reflexive.

The supporting paper cites the source paper as reference 5 and explicitly provides a correction of the proof of this result. Its introduction also lists the same assertion as one of the two basic known complex-case results.

Identification

The hypotheses and conclusion coincide exactly. The source itself states that separability of $C_0(X)$ means metrizability of the one-point compactification. Applying the 2005 theorem with $L = X$ therefore gives an affirmative answer:

$$C_0(X) \text{ separable} \quad \Longrightarrow \quad \mathrm{Iso}(C_0(X)) \text{ algebraically reflexive.}$$

No part of the stated complex-valued question remains open. The supporting paper separately emphasizes that analogous questions for real-valued function spaces are harder; that distinction does not limit the source's complex Banach-algebra question.

Search and provenance

Exact-title, exact-question, and keyword searches on 12 August 2026 located the 2003 Jarosz–Rao finite-codimensional theorem and the decisive corrected 2005 proof above. Because the 2005 paper is a separate publication, cites the source, and states the exact theorem, this packet is classified as `literature_already_answered`, not as an original solution.